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Pickup system and pickup method

Abstract

An aspect of the present invention provides a pickup system which makes, even in a case of transporting a load by using a single unmanned flying object, the unmanned flying object and the load less likely to tilt. A pickup system (1) includes: a holder (11) for holding a load (L); a measuring instrument (14) for measuring the position of the center of gravity of the holder (11) holding the load (L); and a controller (15) for moving, to a position immediately above the position of the center of gravity of the holder (11) holding the load (L), a connector (12) which is attached to the holder (11) in a movable manner and which is for connecting the holder (11) to a drone (D).

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Background/Summary**TECHNICAL FIELD**

(1) The present invention relates to a pickup system and a pickup method, each of which is intended to cause an unmanned flying object to pick up a load.

BACKGROUND ART

(2) Techniques for transporting a load using an unmanned flying object (a drone) have been recently put to practical use. Using an unmanned flying object enables both a reduction in transportation costs and an expansion of the reach of transportation.

(3) In causing an unmanned flying object to pick up a load, it is important what part of the load is connected to the unmanned flying object. If the position of the part connected is inappropriate, the flying object could tilt due to the weight of the load (in a case where the flying object and the load are connected to each other in a fixed manner). This causes problems such as poorer fuel economy and difficulty in controlling a flight route. Alternatively, the load could tilt (in a case where the flying object and the load are connected to each other in an unfixed manner, for example, via a ball joint). This makes the load more likely to be damaged during transportation.

(4) Literatures which disclose techniques for causing an unmanned flying object to pick up a load include, for example, Patent Literature 1. Patent Literature 1 discloses a technique according to which one cargo is transported by using a plurality of drones. The cargo has a plurality of eyebolts attached thereto. Meanwhile, each of the drones has a hook attached thereto. The hook is to be hooked into the eyebolt. In picking up the cargo by using the drones, a control device determines, in accordance with the weight of the cargo and the position of the center of gravity of the cargo, the number of drones to be used and eyebolts to be used among the plurality of eyebolts in order to lift the cargo.

CITATION LIST

Patent Literature

(5) [Patent Literature 1]

(6) Japanese Patent Application Publication Tokukai No. 2020-157922

SUMMARY OF INVENTION

Technical Problem

(7) However, the technique disclosed in Patent Literature 1 is based on the premise that the cargo is picked up by using a plurality of drones. This causes problems of high introduction cost and high operation cost. In addition, in a case of using a drone having a high-capacity lift, it is possible to pick up a cargo by using a single drone. However, the cargo necessarily tilts during transportation since the eyebolts are attached to the cargo in the vicinity of an outer edge of the top plate of the cargo.

(8) An aspect of the present invention has been made in view of the above problems. An object of an aspect of the present invention is to provide a pickup system and a pickup method each of which makes, even in a case of transporting a load by using a single unmanned flying object, the unmanned flying object and the load less likely to tilt.

Solution to Problem

(9) A pickup system in accordance with an aspect of the present invention includes: a holder; a measuring instrument; and a controller. The holder is configured to hold a load. The measuring instrument is configured to measure a position of a center of gravity of the load or the holder holding the load. The controller is configured to move a connector to a position immediately above the position of the center of gravity, the connector being attached to the holder in a movable manner and being for connecting the holder to an unmanned flying object.

(10) A pickup method in accordance with an aspect of the present invention includes: a holding step; a measuring step; and a moving step. The holding step is a step of holding a load by using a holder. The measuring step is a step of measuring, by using a measuring instrument, a position of a center of gravity of the load or the holder holding the load. The moving step is a step of moving, by using a controller, a movable connector to a position immediately above the position of the center of gravity, the movable connector being provided on the holder and being for connecting the holder to an unmanned flying object.

Advantageous Effects of Invention

(11) An aspect of the present invention makes it possible to provide a pickup system and a pickup method each of which makes, even in a case of transporting a load by using a single unmanned flying object, the unmanned flying object and the load less likely to tilt.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view illustrating a configuration of a pickup system in accordance with Embodiment 1 of the present invention.

(2) FIG. 2 is a flowchart illustrating a process flow of a pickup method in accordance with Embodiment 1 of the present invention.

(3) FIG. 3 is a block diagram illustrating a configuration of a computer which functions as a controller included in the pickup system in FIG. 1.

(4) FIG. 4 is a perspective view illustrating a configuration of a pickup system in accordance with Embodiment 2 of the present invention.

(5) FIG. 5 is a plan view of a holder as seen from above which is provided in the pickup system in accordance with Embodiment 2 of the present invention. (a) to (c) of FIG. 5 each illustrate movement of the holder in a moving step included in a pickup method in accordance with Embodiment 2 of the present invention.

(6) FIG. 6 is a flowchart illustrating a process flow of the pickup method in accordance with Embodiment 2 of the present invention.

DESCRIPTION OF EMBODIMENTS

Embodiment 1

(7) (Configuration of Pickup System)

(8) The following description will discuss a pickup system **1** in accordance with Embodiment 1 of the present invention with reference to FIG. **1**. FIG. **1** is a perspective view of a configuration of the pickup system **1**.

(9) The pickup system **1** is a system for causing a drone **D** (an example of an “unmanned flying object” in the claims) to pick up a load **L**. As illustrated in FIG. **1**, the pickup system **1** includes a holder **11**, a connector **12**, a moving mechanism **13**, a measuring instrument **14**, and a controller **15**.

(10) The holder **11** is configured to hold the load **L**. The present embodiment employs, as the holder **11**, a holder composed of one top plate **11a** and two side plates **11b** and **11c**, and two bottom plates **11d** and **11e**.

(11) The top plate **11a**, the two side plates **11b** and **11c**, and the two bottom plates **11d** and **11e** each have a rectangular shape when seen in plan view. In a coordinate system illustrated in FIG. **1**, one side plate **11b** of the two side plates **11b** and **11c** is joined to the top plate **11a** such that an edge of the side plate **11b** which is on a z-axis positive side of the side plate **11b** and parallel to a y axis is joined to an edge of the top plate **11a** which is on an x-axis positive side of the top plate **11a** and parallel to the y axis. The other side plate **11c** is joined to the top plate **11a** such that an edge of the side plate **11c** which is on the z-axis positive side of the side plate **11c** and parallel to the y axis is joined to an edge of the top plate **11a** which is on an x-axis negative side of the top plate **11a** and parallel to the y axis. One bottom plate **11d** of the bottom plates **11d** and **11e** is joined to the side plate **11b** such that an edge of the bottom plate **11d** which is on an x-axis positive side of the bottom plate **11d** and parallel to the y axis is joined to an edge of the side plate **11b** which is on a z-axis negative side of the side plate **11b** and parallel to the y axis. The other bottom plate **11e** is joined to the side plate **11c** such that an edge of the bottom plate **11e** which is on an x-axis negative side of the bottom plate **11e** and parallel to the y axis is joined to an edge of the side plate **11c** which is on a z-axis negative side of the side plate **11c** and parallel to the y axis. The sum of the dimensions of the two bottom plates **11d** and **11e** in the x-axis direction is smaller than the dimension of the top plate **11a** in the x-axis direction. In short, the holder **11** has a shape of a box which has a pair of open side surfaces that face each other and which also has a bottom that has a transverse open center part.

(12) The load **L** is put in the holder **11** through the open side surfaces and/or the center part of the bottom. The two side plates **11b** and **11c** restrict movement of the load **L** in the x-axis positive direction and the x-axis negative direction. Meanwhile, the movement of the load **L** in the z-axis negative direction is restricted by the two bottom plates **11d** and **11e**. The holder **11** can optionally include a clip (not illustrated) attached thereto for restricting the movement of the load **L** in the y-axis positive direction, the y-axis negative direction, and the z-axis positive direction.

(13) The connector **12** is configured to connect the holder **11** to the drone **D**. The moving mechanism **13** is configured to move the connector **12**. The present embodiment employs, as the moving mechanism **13**, a moving mechanism including a first slider **13a** and a second slider **13b**.

(14) The first slider **13a** is a slider which slides on the holder **11** in a first direction. According to the present embodiment, a pair of rails is provided on the top plate **11a** of the holder **11** so as to extend parallel to the x axis. The first slider **13a** is slidably attached to these rails. This allows the first slider **13a** to slide on the holder **11** in the x-axis direction.

(15) The second slider **13b** is a slider which slides on the first slider **13a** in a second direction intersecting the first direction. According to the present embodiment, a pair of rails is provided on the first slider **13a** so as to extend parallel to the y axis. The second slider **13b** is slidably attached to these rails. This allows the second slider **13b** to slide on the first slider **13a** in the y-axis direction.

(16) According to the present embodiment, the connector **12** is fixed to the second slider **13b**. As described above, the second slider **13b** is capable of sliding on the first slider **13a** in the second direction (the y-axis direction in the present embodiment), and the first slider **13a** is capable of

sliding on the holder **11** in the first direction (the x-axis direction in the present embodiment). This allows the connector **12** to freely move, on the holder **11**, within a plane which contains axes in the first direction and the second direction (an x-y plane in the present embodiment).

(17) The measuring instrument **14** is configured to measure the position of the center of gravity of the holder **11** holding the load L. The present embodiment employs a six-axis force sensor as the measuring instrument **14**, although the measuring instrument **14** can be configured to carry out computation using a plurality of load cells. According to the present embodiment, the measuring instrument **14** includes: a plate **14a** on which the holder **11** holding the load L is to be placed; and four six-axis force sensors **14b1** to **14b4** which are installed in a manner that supports the plate **14a**. The six-axis force sensor refers to a sensor capable of detecting force F_x in the x-axis direction, force F_y in the y-axis direction, and force F_z in the z-axis direction, and moment M_x about the x axis, moment M_y about the y axis, and moment M_z about the z axis. The measuring instrument **14** provides, to the controller **15**, sensor signals which are outputted from the four six-axis force sensors **14b1** to **14b4**, respectively. In FIG. 1, the four six-axis force sensors **14b1** to **14b4** are installed on four corners of the plate **14a**, respectively. The six-axis force sensor **14b4** is hidden behind the plate **14a** and not illustrated in FIG. 1.

(18) The controller **15** is configured to calculate, on the basis of the sensor signals obtained from the measuring instrument **14**, the position of the center of gravity of the holder **11** holding the load L, and also to control the moving mechanism **13** so that the connector **12** is located immediately above the calculated position of the center of gravity. The present embodiment employs, as the controller **15**, a computer which includes at least one processor and at least one memory. It is a known method to calculate, on the basis of the sensor signals outputted from the four six-axis force sensors **14b1** to **14b4**, respectively, the position of the center of gravity of an object (the holder **11** holding the load L) placed on the plate **14a**. Therefore, the description of the method is omitted.

(19) It should be noted that a connector **D1** of the drone D and the connector **12** of the holder **11** can be connected to each other either manually by an operator or automatically by the drone D. In the latter case, the controller **15** moves the connector **12** to a position immediately above the calculated position of the center of gravity, and then notify, of the position of the connector **12** thus moved, the drone D or a flight control device (not illustrated; the same applies below) for carrying out flight control of the drone D. This enables the drone D to automatically connect the connector **D1** of the drone D and the connector **12** of the holder **11** to each other.

(20) (Process Flow of Pickup Method)

(21) The following description will discuss a pickup method **S1** in accordance with Embodiment 1 of the present invention with reference to FIG. 2. FIG. 2 is a flowchart illustrating a process flow of the pickup method **S1**.

(22) The pickup method **S1** is a method for causing, by using the pickup system **1**, the drone D to pick up the load L. As illustrated in FIG. 2, the pickup method **S1** includes a holding step **S11**, a measuring step **S12**, and a moving step **S13**.

(23) The holding step **S11** is a step of holding the load L by using the above-described holder **11**. The measuring step **S12** is a step of measuring, by using the above-described measuring instrument **14**, the position of the center of gravity of the holder **11** holding the load L. The moving step **S13** is a step of moving, by using the above-described controller **15**, the connector **12** which is provided on the holder **11** and which is for connecting the holder **11** to the drone D, to a position immediately above the position of the center of gravity of the holder **11** holding the load L.

(24) It should be noted that the connector **D1** of the drone D and the connector **12** of the holder **11** can be connected to each other either manually by an operator or automatically by the drone D. In the latter case, a notifying step can be additionally carried out after the moving step **S13** is carried out. The notifying step is a step of notifying, of the position of the connector **12** moved, the drone D or the flight control device for carrying out flight control of the drone D. In the notifying step, the controller **15** is used to carry out notification. This allows the drone D to automatically connect

the connector **D1** of the drone **D** and the connector **12** of the holder **11** to each other.

(25) (Effects of Pickup System and Pickup Method)

(26) As above, the pickup system **1** in accordance with the present embodiment includes the holder **11** for holding the load **L**, the measuring instrument **14** for measuring the position of the center of gravity of the holder **11** holding the load **L**, and the controller **15** for moving, to a position immediately above the position of the center of gravity of the holder **11** holding the load **L**, the connector **12** which is attached to the holder **11** in a movable manner and which is for connecting the holder **11** to the drone **D**.

(27) The pickup method **S1** in accordance with the present embodiment includes: the holding step **S11** of holding the load **L** by using the holder **11**; the measuring step **S12** of measuring, by using the measuring instrument **14**, the position of the center of gravity of the holder **11** holding the load **L**; and the moving step **S13** of moving, by using the controller **15**, the connector **12** which is provided on the holder **11** and which is for connecting the holder **11** to the drone **D**, to a position immediately above the position of the center of gravity of the holder **11** holding the load **L**.

(28) Thus, when the connector **12** of the holder **11** and the connector **D1** of the drone **D** are connected to each other with use of the pickup system **1** or according to the pickup method **S1**, the position of the center of gravity of the holder **11** holding the load **L** comes to be located immediately below the connector **D1** of the drone **D**. This makes it possible to effectively prevent the drone **D** and the load **L** from tilting. The drone **D** further includes a plurality of propellers (four propellers in the present embodiment). The pickup system **1** and the pickup method **S1** can make respective drive outputs of the plurality of propellers equal to each other. This makes it possible to reduce a power consumption of the drone **D**. In other words, it is possible to increase the drive efficiency of the drone **D**.

(29) According to the present embodiment, the measuring instrument **14** includes: the plate **14a** on which the holder **11** holding the load **L** is to be placed; and at least one six-axis force sensor (e.g., six-axis force sensors **14b1** to **14b4**) which is installed in a manner that supports the plate **14a**.

(30) This enables accurate measurement of the position of the center of gravity of the holder **11** holding the load **L**.

(31) In particular, according to the present embodiment, the measuring instrument **14** includes: the plate **14a** on which the holder **11** holding the load **L** is to be placed; and four six-axis force sensors **14b1** to **14b4** which are installed in a manner that supports the plate **14a**.

(32) This enables more accurate measurement of the position of the center of gravity of the holder **11** holding the load **L**.

(33) According to the present embodiment, the pickup system **1** further includes the moving mechanism **13** for moving the connector **12**. The moving mechanism **13** includes the first slider **13a** which slides on the holder **11** in the first direction and the second slider **13b** which slides on the first slider **13a** in the second direction intersecting the first direction. The connector **12** is fixed to the second slider **13b**.

(34) This makes it possible to easily move the connector **12** to a position immediately above the position of the center of gravity of the holder **11** holding the load **L**.

(35) According to the present embodiment, the controller **15** moves the connector **12** to the position immediately above the position of the center of gravity of the holder **11** holding the load **L**, and then notifies, of the position of the connector **12**, the drone **D** or the flight control device for carrying out flight control of the drone **D**.

(36) This makes it possible for the drone **D** to automatically connect the connector **D1** of the drone **D** and the connector **12** of the holder **11** to each other.

(37) It should be noted that although the present embodiment employs the configuration in which the position of the center of gravity of the holder **11** holding the load **L** is measured by using the measuring instrument **14**, the present invention is not limited to this. An embodiment of the present invention can employ a configuration in which the position of the center of gravity of the load **L** is

measured by using the measuring instrument **14**. In this case, the controller **15** moves the connector **12** to a position immediately above the position of the center of gravity of the load **L**.

(38) In a case where the weight of the holder **11** is sufficiently smaller than the weight of the load **L**, the position of the center of gravity of the load **L** substantially coincides with the position of the center of gravity of the holder **11** holding the load **L**. Accordingly, even in a case of employing the configuration in which the position of the center of gravity of the load **L** is measured, it is possible to achieve an effect which is substantially the same as that achieved in the case of employing the configuration in which the position of the center of gravity of the holder **11** holding the load **L** is measured.

(39) (Software Implementation of Controller)

(40) Some or all of the foregoing functions of the controller **15** can be implemented by hardware such as an integrated circuit (IC chip) or by software. In the latter case, the functions of the controller **15** are each implemented by, for example, a computer which executes instructions of a program **P** that is software.

(41) FIG. **3** illustrates an example of such a computer (hereinafter, referred to as computer **C**). As illustrated in FIG. **3**, the computer **C** includes at least one processor **C1** and at least one memory **C2**. The memory **C2** stores a program **P** for causing the computer **C** to operate as the controller **15**. In the computer **C**, the processor **C1** reads the program **P** from the memory **C2** and executes the program, so that the functions of the controller **15** are implemented.

(42) Examples of the processor **C1** encompass a central processing unit (CPU), a graphic processing unit (GPU), a digital signal processor (DSP), a micro processing unit (MPU), a floating point number processing unit (FPU), a physics processing unit (PU), a micro controller, and a combination thereof. Examples of the memory **C2** encompass a flash memory, a hard disk drive (HDD), a solid state drive (SSD), and a combination thereof.

(43) Note that the computer **C** can further include a random access memory (RAM), in which the program **P** is loaded when the program **P** is to be executed and in which various kinds of data are temporarily stored. The computer **C** can further include a communication interface through which data can be transmitted to and received from another device. The computer **C** can further include an input-output interface through which an input device such as a keyboard and a mouse and/or an output device such as a display and a printer is/are connected to the computer **C**.

(44) The program **P** can be stored in a non-transitory, tangible storage medium **M** capable of being read by the computer **C**. Examples of such a storage medium **M** encompass a tape, a disk, a card, a semiconductor memory, and a programmable logic circuit. The computer **C** can obtain the program **P** via the storage medium **M**. Alternatively, the program **P** can be transmitted via a transmission medium. Examples of such a transmission medium encompass a communication network and a broadcast wave. The computer **C** can also obtain the program **P** via the transmission medium.

Embodiment 2

(45) (Configuration of Pickup System)

(46) The following description will discuss a pickup system **2** in accordance with Embodiment 2 of the present invention with reference to FIGS. **4** and **5**. FIG. **4** is a perspective view of a configuration of the pickup system **2**. FIG. **5** is a plan view of a holder **21** seen from above (from a z-axis positive side), the holder **21** being provided in the pickup system **2**. (a) to (c) of FIG. **5** each illustrate movement in a moving step **S24** included in a pickup method in accordance with Embodiment 2 of the present invention.

(47) The pickup system **2**, like the pickup system **1**, is a system for causing a drone **D** (an example of an “unmanned flying object” in the claims) to pick up a load **L**. As illustrated in FIG. **4**, the pickup system **2** includes a holder **21**, a connector **22**, a moving mechanism **23**, a connector **24**, a measuring instrument **25**, a cage **26**, and an actuator group **27**. The pickup system **2** also includes a controller (illustration of which is omitted in FIG. **4**).

(48) The pickup system **1** in accordance with Embodiment 1 employs a configuration in which the

holder **11** is placed on the measuring instrument **14** as illustrated in FIG. **1**. On the other hand, in the pickup system **2**, the holder **21** is fixed to a top plate **26a** of the cage **26** via the connector **22**, the connector **24**, and the measuring instrument **25**, so as to hang from the top plate **26a**, as illustrated in FIG. **4**. Therefore, in the present embodiment, first, the cage **26** and the actuator group **27** will be discussed, and then, the holder **21**, the connector **22**, the moving mechanism **23**, the connector **24**, and the measuring instrument **25** will be discussed.

(49) The cage **26**, which is an example of a support, includes the top plate **26a**, four columns **26b1**, **26b2**, **26b3**, and **26b4** and a frame **26c**. The frame **26c** is composed of four beams **26c1**, **26c2**, **26c3**, and **26c4**.

(50) The top plate **26a** is a plate member which has a rectangular shape when either one of a pair of main surfaces of the top plate **26a** is seen in plan view. One of the pair of main surfaces (lower main surface in FIG. **4**) of the top plate **26a** is provided, in the center of the one main surface, with a placement area on which the connector **24** is to be placed via the measuring instrument **25**. The connector **24** and the measuring instrument **25** will be discussed later. The top plate **26a** preferably has high rigidity so that even in a case where the weight of the load **L** is large, deformation such as bending is unlikely to occur. In the present embodiment, a metal plate is used as the top plate **26a**.

(51) At respective four corners of the top plate **26a** having a rectangular shape, the columns **26b1**, **26b2**, **26b3**, and **26b4** are fixed to the lower main surface of the top plate **26a** such that a direction in which the columns **26b1**, **26b2**, **26b3**, and **26b4** extend is orthogonal to the lower main surface. That is, the columns **26b1**, **26b2**, **26b3**, and **26b4** serve as leg parts which support the top plate **26a**. The columns **26b1**, **26b2**, **26b3**, and **26b4** have lengths equal to each other. Note that the columns **26b1**, **26b2**, **26b3**, and **26b4** each may be provided, at its lower end, a screw-type adjustment mechanism for adjusting the length of each of the leg parts. The top plate **26a** can be fixed such that the placement area described above is substantially horizontal (more preferably, horizontal), by using such an adjustment mechanism.

(52) The frame **26c** is provided at intermediate parts of the columns **26b1**, **26b2**, **26b3**, and **26b4**. The frame **26c** is composed of the beams **26c1**, **26c2**, **26c3**, and **26c4** each of which connects between adjacent columns. The beam **26c1** connects the columns **26b1** and **26b2**. The beam **26c2** connects the columns **26b2** and **26b3**. The beam **26c3** connects the columns **26b3** and **26b4**. The beam **26c4** connects the columns **26b4** and **26b1**. The frame **26c** configured as above surrounds a bottom plate **21d** of the holder **21** in a case where the holder **21** is connected to the connector **24** via the connector **22** and the moving mechanism **23**.

(53) The frame **26c** is provided with the actuator group **27** which is composed of actuators **271**, **272**, **273**, and **274**. The actuator group **27** is an example of the adjustment mechanism. Each of the actuators **271**, **272**, **273**, and **274** is a shaft-like member which linearly extends, and is configured such that an axial length, which is the length along a center axis of each of the actuators **271**, **272**, **273**, and **274**, can be adjusted to a desired length. In the present embodiment, each of the actuators **271**, **272**, **273**, and **274** is an air actuator which uses air pressure for adjusting the axial length. Note however that a source of power for use in adjusting the axial length is not limited to air pressure but can be oil pressure or electric power.

(54) The actuator **271** is fixed to a middle point of the beam **26c1** such that the center axis of the actuator **271** is orthogonal to a direction in which the beam **26c1** extends. Similarly, the actuator **272** is fixed to a middle point of the beam **26c2** such that the center axis of the actuator **272** is orthogonal to a direction in which the beam **26c2** extends. Similarly, the actuator **273** is fixed to a middle point of the beam **26c3** such that the center axis of the actuator **273** is orthogonal to a direction in which the beam **26c3** extends. Similarly, the actuator **274** is fixed to a middle point of the beam **26c4** such that the center axis of the actuator **274** is orthogonal to a direction in which the beam **26c4** extends. Each of the actuator **271**, **272**, **273**, and **274** is provided such that a movable tip end portion of the actuator protrudes inward on an inner side of the frame **26c**.

(55) The actuator group **27** configured as above can restrict movement of the holder **21** in a y-axis

direction and move the holder **21**, by controlling the axial lengths of the actuators **271** and **273** which are opposed to each other. Similarly, the actuator group **27** can restrict movement of the holder **21** in an x-axis direction and move the holder **21**, by controlling the axial lengths of the actuators **272** and **274** which are opposed to each other. In the present embodiment, the holder **21** is moved within a horizontal plane by using the actuator group **27** so that the center **C25** (see FIG. 6) of a six-axis force sensor **25** corresponding to the position of the connector **22** is moved to a position immediately above the position of the center of gravity **CL** (see FIG. 5) of the holder **21** holding the load **L**. Note that movement of the actuator group **27** is controlled by the controller discussed later.

(56) Next, the following will discuss the holder **21**, the connector **22**, the moving mechanism **23**, the connector **24**, and the measuring instrument **25**.

(57) The holder **21** is configured to hold the load **L** in the same manner as the holder **11**. The present embodiment employs, as the holder **21**, a holder composed of one top plate **21a**, columns **21b1**, **21b2**, **21b3**, and **21b4**, and one bottom plate **21d**.

(58) The top plate **21a** and the bottom plate **21d** each have a rectangular shape when seen in plan view. The top plate **21a** and the bottom plate **21d** have shapes congruent to each other when seen in plan view. In addition, the four columns **21b1** to **21b4** having lengths equal to each other are present between the top plate **21a** and the bottom plate **21d**. Therefore, the distance between the top plate **21a** and the bottom plate **21d** is equal to the lengths of the columns **21b1** to **21b4**. The columns **21b1** to **21b4** are provided at the respective four corners of the top plate **21a** and the bottom plate **21d**. The holder **21** configured as described above has a shape of a box which has two pairs of open side surfaces that face each other. In a coordinate system illustrated in FIG. 4, the holder **21** is placed such that the top plate **21a** and the bottom plate **21d**, each of which have a rectangular shape, have long sides parallel to the x axis, and short sides parallel to the y axis, and such that the direction in which the columns **21b1** to **21b4** extend is parallel to a z axis. Note that the coordinate system illustrated in FIG. 4 is set such that an x-y plane is horizontal and the z axis is parallel to a vertical direction.

(59) In the present embodiment, the load **L** is fixed to the bottom plate **21d** of the holder **21** by using a band-like sling. Alternatively, the load **L** can be fixed to the bottom plate **21d** of the holder **21** by using Velcro (registered trademark). Alternatively, it is possible to use a sling and Velcro in combination in order to fix the load **L** to the bottom plate **21d** of the holder **21**. In this manner, the holder **21** holds the load **L**.

(60) The connector **22** is an example of a first connector and is configured to connect the holder **21** to the drone **D** in the same manner as the connector **12**. The connector **22** is configured in the same manner as the connector **12**, and therefore an explanation thereof is omitted here.

(61) The moving mechanism **23** is configured to move the connector **22** within a horizontal plane (the x-y plane) that is a plane which contains axes in a first direction (the x-axis direction) and a second direction (the y-axis direction), in the same manner as the moving mechanism **13**. In the present embodiment, the moving mechanism **23** is a moving mechanism which includes a first slider **23a**, a second slider **23b**, a first guide **23c**, a second guide **23d**, a first timing belt **23e**, a second timing belt **23f**, first timing belt pulleys **23g**, and second timing belt pulleys **23h**.

(62) The first slider **23a** and the second slider **23b** are configured in the same manner as the first slider **13a** and the second slider **13b**, respectively. Therefore, in the present embodiment, explanations of the first slider **23a** and the second slider **23b** are omitted.

(63) The first guide **23c** is present between the first slider **23a** and a pair of rails provided on the top plate **21a** of the holder **21**, and serves as a guide which restricts movement of the first slider **23a** relative to the pair of rails, to the x-axis direction which is the first direction. In order to restrict the movement of the first slider **23a**, the pair of rails extends in parallel to the x-axis direction.

(64) The second guide **23d** is present between the first slider **23a** and the second slider **23b**, and serves as a guide which restricts movement of the second slider **23b** relative to the first slider **23a**,

to the y-axis direction which is the second direction.

(65) The first timing belt **23e** and the two first timing belt pulleys **23g** are configured to switch a state of the first slider **23a** between a slidable state and a non-slidable state. In other words, the first timing belt **23e** and the two first timing belt pulleys **23g** are an example of a first switch. The two first timing belt pulleys **23g** are pulleys provided on a straight line which is parallel to each of the pair of rails and which is at equal distances from the pair of rails. The first timing belt **23e** is placed on the two first timing belt pulleys **23g** provided as described above, so that the first timing belt **23e** extends in a direction parallel to the x-axis direction. At least one of the two first timing belt pulleys **23g** is provided with an electromagnetic clutch for switching a state of the pulleys between a rotatable state and a non-rotatable state. This electromagnetic clutch is controlled by the controller. Further, the first timing belt **23e** having an annular shape is stretched between the two first timing belt pulleys **23a**. A portion of the first slider **23a** is fitted to a portion of the first timing belt **23e**. Therefore, in a case where the first timing belt pulley **23g** is in the rotatable state, the first slider **23a** is in the slidable state. On the other hand, in a case where the first timing belt pulley **23g** is in the non-rotatable state, the first slider **23a** is in the non-slidable state. In this configuration, the first slider **23a** is caused to be in the slidable state by turning on the electromagnetic clutch provided in the first timing belt pulley **23g**, and the first slider **23a** is caused to be in the non-slidable state by turning off the electromagnetic clutch.

(66) The second timing belt **23f** and the two second timing belt pulleys **23h** are configured to switch a state of the second slider **23b** between a slidable state and a non-slidable state. The second timing belt **23f** and the two second timing belt pulleys **23h** are an example of a second switch. The two second timing belt pulleys **23h** are each a pulley provided on each of the pair of rails. The second timing belt **23f** is placed on the two second timing belt pulleys **23h** provided as described above, so that the second timing belt **23f** extends in a direction parallel to the y-axis direction. The second timing belt **23f** and the two second timing belt pulleys **23h** are configured in the same manner as the first timing belt **23e** and the two first timing belt pulleys **23g**, except that the second timing belt **23f** is stretched not in the x-axis direction but in the y-axis direction. In this configuration, the second slider **23b** is caused to be in the slidable state by turning on the electromagnetic clutch provided in the second timing belt pulleys **23h**, and the second slider **23b** is caused to be in the non-slidable state by turning off the electromagnetic clutch.

(67) As described above, in the present embodiment, in order to switch the state of each of the first slider **23a** and the second slider **23b** between the slidable state and the non-slidable state, that is, in order to switch the state of each of the first timing belt **23e** and the second timing belt **23f** between a drivable state and a non-drivable state, the first timing belt pulley **23g** and the second timing belt pulley **23h** each includes the electromagnetic clutch. Note however that the first timing belt pulley **23g** and the second timing belt pulley **23h** each may include a mechanical clutch or a mechanical brake in place of the electromagnetic clutch.

(68) The connector **24** is an example of a second connector and is connected to the connector **22**. The connector **24** corresponds to the connector **22**, and is a second connector. The connector **24** makes a pair of connectors with the connector **22**. In the present embodiment, a quick coupling is used as the connectors **22** and **24**. Therefore, the connector **22** can be attached to or detached from the connector **24** with a single-touch action. The connector **24** is configured in the same manner as a connector which the drone D is provided with.

(69) The measuring instrument **25** is provided at a portion of the connector **24** on a side opposite to the connector **22**. The present embodiment employs a six-axis force sensor as the measuring instrument **25**. The six-axis force sensor can detect force and moment along/about three axes of an orthogonal coordinate system. In the pickup system **2**, the six-axis force sensor is placed such that among the three axes of the six-axis force sensor, a first axis is parallel to the x axis, a second axis is parallel to the y axis, and a third axis is parallel to the z axis. In other words, the six-axis force sensor is a sensor capable of detecting force F_x in the x-axis direction, force F_y in the y-axis

direction, and force F_z in the z-axis direction, and moment M_x about the x axis, moment M_y about the y axis, and moment M_z about the z axis. Since the present embodiment employs a single six-axis force sensor as the measuring instrument 25, the measuring instrument 25 is also referred to as the six-axis force sensor 25.

(70) The controller (not illustrated in FIG. 4) is configured to measure, on the basis of the sensor signals obtained from the six-axis force sensor 25, the position of the center of gravity CL (see FIG. 6) of the holder 21 holding the load L, with reference to the center C25 (see FIG. 6) of the six-axis force sensor 25 corresponding to the position of the connector 22. The controller is configured to also control the moving mechanism 23 so that the center C25 is located immediately above the position of the center of gravity CL.

(71) The controller moves the center C25 to a position immediately above the position of the center of gravity CL, by moving the connector 22 relative to the holder 21, so that among the three forces F_x , F_y , and F_z and the three moments M_x , M_y , and M_z , the three moments, which are detected by the six-axis force sensor, are not more than a predetermined value (preferably, so that the three moments are zero). Note that in order to move the connector 22 relative to the holder 21, the controller controls the actuator group 27. The actuator group 27 will be described in detail later.

(72) As illustrated in FIG. 4, the holder 21 is fixed to the top plate 26a of the cage 26, via the connector 22, the connector 24, and the six-axis force sensor 25. The holder 21 is fixed so as to hang from the top plate 26a. Therefore, the moment M_z out of the three moments M_x , M_y , and M_z is zero, regardless of whether or not the center C25 is located immediately above the position of the center of gravity CL. On the other hand, the moments M_x and M_y out of the three moments M_x , M_y , and M_z are respective moments about the x axis and the y axis contained in a horizontal plane. In a case where the center C25 is located immediately above the position of the center of gravity CL, both of the moments M_x and M_y are zero. On the other hand, in a case where the center C25 is not located immediately above the position of the center of gravity CL, at least one of the moments M_x and M_y is not zero. Therefore, the controller causes the position of the center of gravity CL and the position of the center C25 coincide with each other, by moving the holder 21, so that both of the moment M_x and the moment M_y are not more than a predetermined value.

(73) In the present embodiment, the controller (illustration of which is omitted in FIG. 4) is configured in the same manner as the controller 15. In other words, the controller can be realized by hardware provided in an integrated circuit (IC chip) or the like or can be alternatively realized by software, in the same manner as the controller 15. In the latter case, the functions of the controller are each implemented by, for example, a computer (e.g., computer C illustrated in FIG. 3) which executes instructions of a program P that is software.

(74) (Process Flow of Pickup Method)

(75) The following description will discuss a pickup method S2 in accordance with Embodiment 2 of the present invention with reference to FIGS. 5 and 6. FIG. 5 is a plan view of the holder 21 as seen from above (from the z-axis positive side) which is provided in the pickup system 2 as described above. (a) to (c) of FIG. 5 each illustrate movement in the moving step S24 included in the pickup method in accordance with Embodiment 2 of the present invention. FIG. 6 is a flowchart illustrating a process flow of the pickup method in accordance with the present embodiment.

(76) The pickup method S2 is a method for causing, by using the pickup system 2, the drone D to pick up the load L, in the same manner as the pickup method S1. As illustrated in FIG. 6, the pickup method S2 includes a holding step S21, a fixing step S22, a measuring step S23, and the moving step S24. The holding step S21 is a step corresponding to the holding step S11, the measuring step S23 is a step corresponding to the measuring step S12, and the moving step 24 is a step corresponding to the moving step S13.

(77) The holding step S21 is a step of holding the load L by using the above-described holder 21. In the holding step S21, the load L is fixed to the bottom plate 21d of the holder 21 by using, for

example, a band-like sling. At this time, the load L is not necessarily fixed such that the center of gravity of the load L coincides with the center of gravity of the holder **21**.

(78) The fixing step **S22** is a step which is carried out after the holding step **S21**. The fixing step **S22** is a step of fixing the holder **21** where the load L is set, to the top plate **26a** of the cage **26** via the connector **22**, the connector **24**, and the six-axis force sensor **25**. (a) of FIG. 5 illustrates a state in which the holder **21** is fixed to the top plate **26a**. Note however that in FIG. 5, illustration of the top plate **26a** is omitted. In (a) of FIG. 5, the center **C25** of the six-axis force sensor **25** is set at the center of the top plate **26a**. In this case, the center of the top plate **26a** is a reference position, and the center **C25** corresponds to the center of the connector **22** and the center of the connector **24**. The top plate **26a** is fixed such that the placement area is substantially horizontal (more preferably, horizontal). Further, the six-axis force sensor **25** is placed so that the placement area is parallel to a plane which contains a first axis (an axis parallel to the x axis) and a second axis (an axis parallel to the y axis) among the three axes along/about which force and moment are to be detected. Therefore, the six-axis force sensor **25** is placed so as to be substantially horizontal to the plane which contains the first axis and the second axis. Further, in (a) of FIG. 5, the center of gravity CL of the holder **21** holding the load L is located in a region on the x-axis positive side and on the y-axis negative side.

(79) The measuring step **S23** is a step of measuring, by using the above-described six-axis force sensor **25**, the position of the center of gravity CL. Specifically, the controller measures the position of the center of gravity CL, with reference to two moment signals representing three moments M_x and M_y among sensor signals outputted from the six-axis force sensor **25**. The moment M_x is an example of moment about the first axis, and the moment M_y is an example of moment about the second axis.

(80) The moving step **S24** is a step of moving, by using the above-described controller, the center **C25** to a position immediately above the position of the center of gravity CL. The controller moves the holder **21** by using the actuator group **27**, in accordance with the position of the center of gravity CL which has been measured in the measuring step **S23**. The arrow **F27** shown in (a) of FIG. 5 indicates a direction of force which acts on the holder **21** when the holder **21** is moved toward the y-axis positive side with use of the actuators **272** and **274**. The arrow **F27** shown in (b) of FIG. 5 indicates a direction of force which acts on the holder **21** when the holder **21** is moved toward the x-axis negative side with use of the actuators **271** and **273**.

(81) As described above, the center **C25** can be moved to the position immediately above the position of the center of gravity CL, by carrying out the pickup method **S2**.

(82) Note that the pickup method **S2** can be configured to carry out the measuring step **S23** again after the moving step **S24**. The controller measures the position of the center of gravity CL, with reference to three moment signals representing the three moments M_x , M_y , and M_z . In a case where the three moments M_x , M_y , and M_z are not more than the predetermined value, the controller should determine that the position of the center **C25** is at the position immediately above the center of gravity CL, and end the pickup method **S2**. On the other hand, in a case where any one of the three moments M_x , M_y , and M_z is more than the predetermined value, the controller should carry out the moving step **S24** again. As described above, the pickup method **S2** may be configured to repeatedly carry out the measuring step **S23** and the moving step **S24** until the three moments M_x , M_y , and M_z are not more than the predetermined value.

(83) (Effects of Pickup System and Pickup Method)

(84) As described above, the pickup system **2** in accordance with the present embodiment yields the same effect as the pickup system **1**. Further, the pickup method **S2** in accordance with the present embodiment yields the same effect as the pickup method **S1**. The following will describe effects which are yielded by the pickup system **2** and the pickup method **2** in addition to the above effects.

(85) The image pickup system **2** further includes: the moving mechanism **23** which includes the first slider **23a** and the second slider **23b**; the second connector (the connector **24** in the present

embodiment); the support **26** provided so that the placement area, on which the second connector (the connector **24**) is placed, becomes substantially horizontal; the six-axis force sensor **25**; and the actuator group **27**. In the pickup system **2**, the first connector (the connector **22** in the present embodiment) is fixed to the second slider **23b**.

(86) The above configuration makes it possible to realize the pickup system **2** which is simpler than the pickup system **1**, since the pickup system **2** can be configured by using the single six-axis force sensor **25**.

(87) Further, the pickup system **2** further includes the first switch (in the present embodiment, the first timing belt **23e** and the two first timing belt pulleys **23g**) and the second switch (in the present embodiment, the second timing belt **23f** and the two second timing belt pulleys **23h**).

(88) According to the above configuration, after the holder **21** is caused to hold the load L, the first slider **23a** and the second slider **23b** are set to the slidable state while the position of the connector **22** is being adjusted relative to the position of the center of gravity CL of the holder **21** holding the load L. After this adjustment, the first slider **23a** and the second slider **23b** can be set to the non-slidable state. This makes it possible to prevent displacement of the connector **22** relative to the position of the center of gravity CL of the holder **21** holding the load L while the drone D is carrying the load L.

(89) Meanwhile, the pickup method S2 further includes the fixing step S22 which is to be carried out after the holding step S21. In the fixing step S22, the six-axis force sensor **25** is placed so that the plane which contains the first axis and the second axis is substantially horizontal, the first axis and the second axis being among the three axes along/about which force and moment are to be detected. Further, in the moving step S24, the connector **22** is moved so that the moment about the first axis and the moment about the second axis are not more than the predetermined value, with reference to a first moment signal indicating the moment about the first axis and a second moment signal about the second axis among the sensor signals outputted from the six-axis force sensor **25**.

(90) The above configuration makes it possible to realize the pickup method S2 which is simpler than the pickup method S1, since the pickup method S2 can be configured by using the single six-axis force sensor **25** as the measuring instrument.

(91) (Supplementary Note)

(92) The present invention is not limited to the above embodiments, but can be altered by a skilled person in the art within the scope of the claims. For example, the present invention also encompasses, in its technical scope, any embodiment derived by combining technical means disclosed in differing embodiments.

(93) TABLE-US-00001 Reference Signs List 1, 2 Pickup system 11, 21 Holder 12, 22 Connector 13, 23 Moving mechanism 14 Measuring instrument 14a Plate 14b Six-axis force sensor 15 Controller 24 Connector 25 Six-axis force sensor 26 Support 26a Top plate (provided with placement area) S1, S2 Pickup method S11, S21 Holding step S22 Fixing step S12, S23 Measuring step S13, S24 Moving step D Drone L Load

Claims

1. A pickup system comprising: a holder for holding a load; a measuring instrument for measuring a position of a center of gravity of the load or the holder holding the load; a connector being attached to the holder in a movable manner and being for connecting the holder to an unmanned flying object; and a controller for moving the connector to a position immediately above the position of the center of gravity, wherein the measuring instrument includes: a plate on which the load or the holder holding the load is to be placed; and at least one six-axis force sensor which is installed in a manner that supports the plate.
2. The pickup system according to claim 1, wherein the measuring instrument includes four of the six-axis force sensors installed in a manner that supports the plate.

3. The pickup system according to claim 1, further comprising a moving mechanism for moving the connector, the moving mechanism including: a first slider configured to slide on the holder in a first direction; and a second slider configured to slide on the first slider in a second direction intersecting the first direction, the connector being fixed to the second slider.
 4. The pickup system according to claim 1, wherein the controller is configured to move the connector to the position immediately above the position of the center of gravity, and then notify the unmanned flying object or a flight control device of a position of the connector, the flight control device being for carrying out flight control of the unmanned flying object.
 5. The pickup system according to claim 1, further comprising: a moving mechanism for moving the connector, the moving mechanism including: a first slider configured to slide on the holder in a first direction; and a second slider configured to slide on the first slider in a second direction intersecting the first direction, the connector being fixed to the second slider; a second connector configured to connect to the connector which serves as a first connector; a support provided so as to make a placement area substantially horizontal, the placement area being an area where the second connector is to be placed; a six-axis force sensor present between the second connector and the placement area; and an adjustment mechanism configured to adjust a position of the holder within a plane which contains axes in the first direction and the second direction, in a state in which the second connector is connected to the first connector, the first connector being fixed to the second slider.
 6. The pickup system according to claim 5, further comprising: a first switch configured to switch a state of the first slider between a slidable state and a non-slidable state; and a second switch configured to switch a state of the second slider between a slidable state and a non-slidable state.
 7. A pickup method comprising the steps of: holding a load by using a holder; fixing the holder to a support via a connector and a six-axis force sensor, measuring, by using a measuring instrument, a position of a center of gravity of the load or the holder holding the load; and moving, by using a controller, a connector to a position immediately above the position of the center of gravity, the connector being provided on the holder and being for connecting the holder to an unmanned flying object, wherein the step of fixing, the six-axis force sensor being placed such that a plane which contains a first axis and a second axis is substantially horizontal, the first axis and the second axis being among three axes along which force is to be detected and about which moment is to be detected, and in the step of moving, the connector being moved so that the moment about the first axis and the moment about the second axis are not more than a predetermined value, with reference to a first moment signal indicating the moment about the first axis and a second moment signal about the second axis among sensor signals outputted from the six-axis force sensor.
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